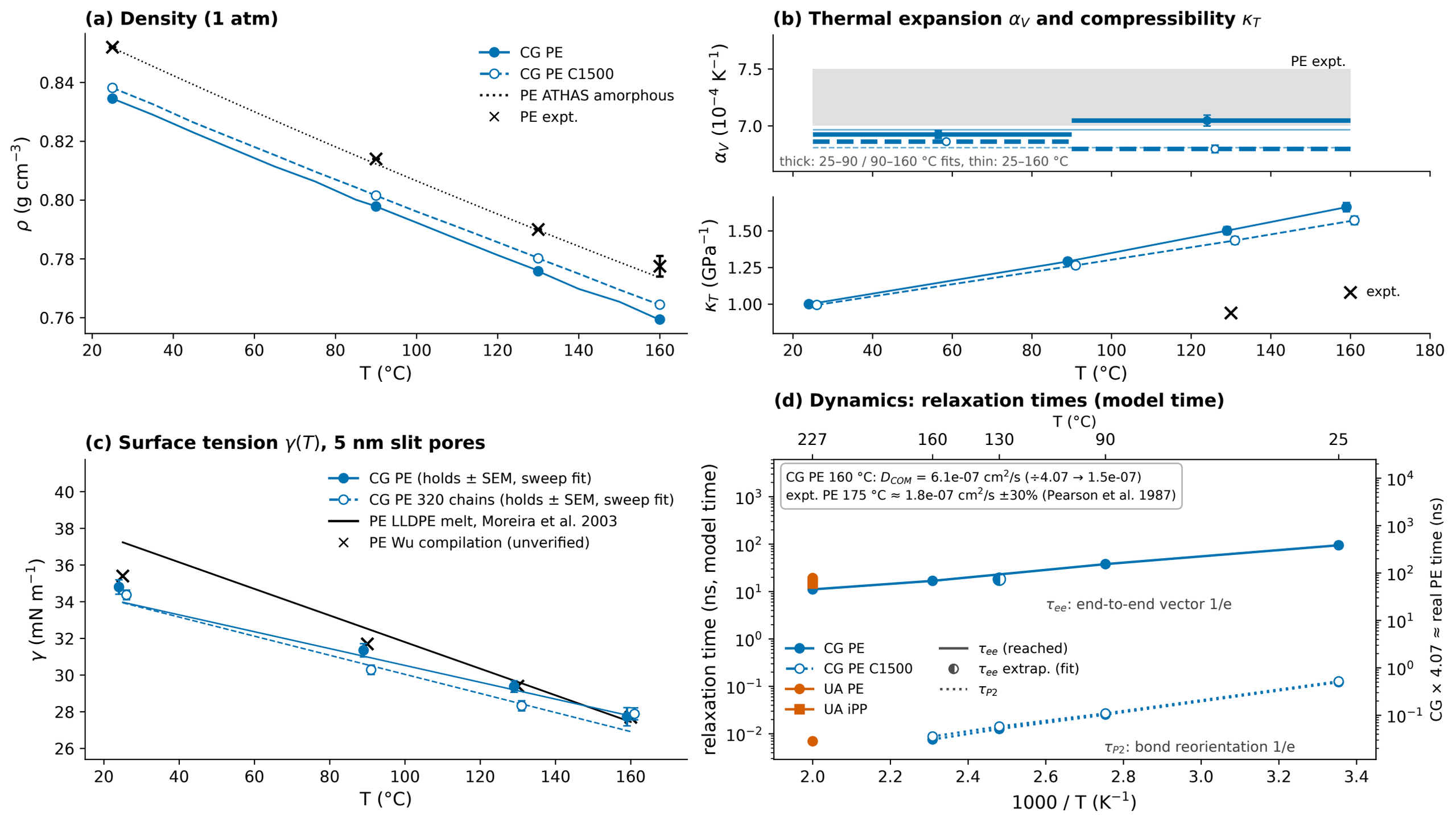
(a) Density (1 atm)
CG PE
CG PE C1500
PE ATHAS amorphous
PE expt.
ρ (g cm⁻³)
T (°C)
(b) Thermal expansion α_V and compressibility κ_T
PE expt.
thick: 25–90 / 90–160 °C fits, thin: 25–160 °C
α_V (10^{-4} K^{-1})
κ_T (GPa^{-1})
expt.
T (°C)
(c) Surface tension $\gamma(T)$, 5 nm slit pores
CG PE (holds ± SEM, sweep fit)
CG PE 320 chains (holds ± SEM, sweep fit)
PE LLDPE melt, Moreira et al. 2003
PE Wu compilation (unverified)
γ (mN m⁻¹)
T (°C)
(d) Dynamics: relaxation times (model time)
T (°C)
CG PE 160 °C: D_{COM} = 6.1e-07 cm²/s (÷4.07 → 1.5e-07)
expt. PE 175 °C ≈ 1.8e-07 cm²/s ±30% (Pearson et al. 1987)
τ_{ee}: end-to-end vector 1/e
τ_{P2}: bond reorientation 1/e
CG PE
CG PE C1500
UA PE
UA iPP
τ_{ee} (reached)
τ_{ee} extrap. (fit)
τ_{P2}
relaxation time (ns, model time)
CG × 4.07 ≈ real PE time (ns)
1000 / T (K⁻¹)